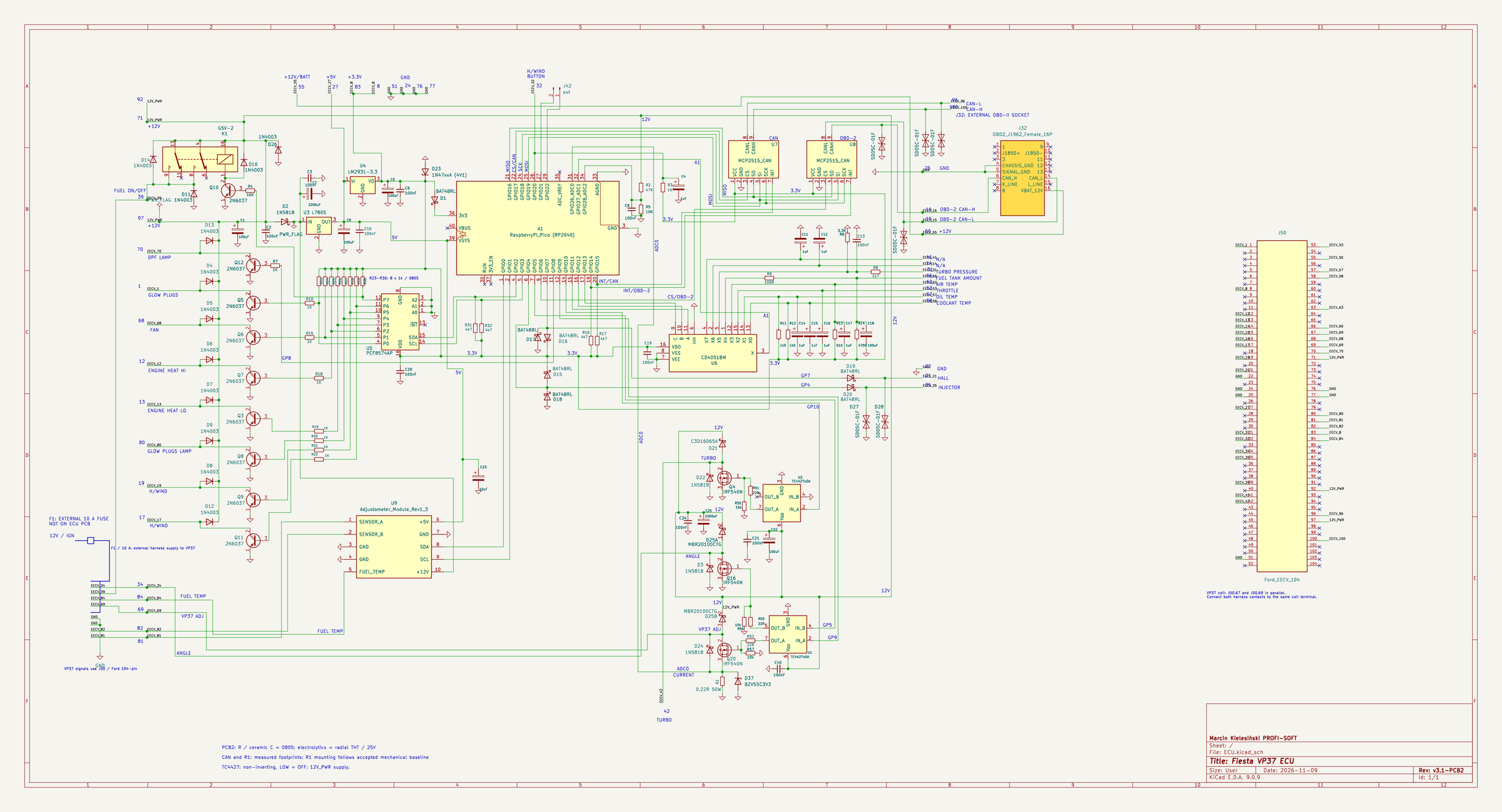

+12V/BATT
+5V
+3.3V
GND
H/WIND BUTTON
J42 ext
12V_PWR
G5V-2 K1
1N4003
FUEL ON/OFF
Q10 2N6037
U3 L7805
PWR_FLAG
U4 LM2931-3.3
A1 RaspberryPi_Pico (RP2040)
BAT48RL
U5 PCF8574AP
R23–R30: 8 x 1k / 0805
DPF LAMP
GLOW PLUGS
FAN
ENGINE HEAT HI
ENGINE HEAT LO
GLOW PLUGS LAMP
H/WIND
U9 Adjustometer_Module_Rev1_3
SENSOR_A
SENSOR_B
FUEL_TEMP
+12V
SDA
SCL
F1: EXTERNAL 10 A FUSE NOT ON ECU PCB
12V / IGN
F1 / 10 A: external harness supply to VP37
VP37 signals use J50 / Ford 104-pin
FUEL TEMP
VP37 ADJ
ANGLE
U7 MCP2515_CAN
U8 MCP2515_CAN
OBD-2
CAN-L
CAN-H
J32: EXTERNAL OBD-II SOCKET
J32 OBD2_J1962_Female_16P
J1850+
J1850-
CHASSIS_GND
SIGNAL_GND
CAN_H
CAN_L
K_LINE
L_LINE
VBAT_12V
OBD-2 CAN-H
OBD-2 CAN-L
+12V
INT/OBD-2
CS/OBD-2
MOSI
MISO
ADCO
U6 CD4051BM
N/A
N/A
TURBO PRESSURE
FUEL TANK AMOUNT
AIR TEMP
THROTTLE
OIL TEMP
COOLANT TEMP
HALL
INJECTOR
D19 BAT48RL
D20 BAT48RL
D27 SD05C-01F
D28 SD05C-01F
C3D16065A
D21
TURBO
D22 1N5818
Q4 IRF540N
U2 TC4427xOA
D25A MBR20100CTG
ANGLE
D3 1N5818
Q16 IRF540N
D25B MBR20100CTG
VP37 ADJ
D24 1N5818
Q20 IRF540N
U1 TC4427xOA
ADCO CURRENT
D37 BZV55C3V3
0.22R 50W
TURBO
J50
Ford_EECV_104
VP37 coil: J50.67 and J50.69 in parallel.
Connect both harness contacts to the same coil terminal.
PCB2: R / ceramic C = 0805; electrolytics = radial THT / 25V
CAN and R1: measured footprints; R1 mounting follows accepted mechanical baseline
TC4427: non-inverting, LOW = OFF; 12V_PWR supply.
Marcin Kielesiński PROFI-SOFT
Sheet: /
File: ECU.kicad_sch
Title: Fiesta VP37 ECU
Size: User
Date: 2026-11-09
Rev: v3.1-PCB2
KiCad E.D.A. 9.0.9
Id: 1/1